

We thank the referees for their comments and positive feedback. We address their concerns below.

Referee A

The manuscript by Gunnink and Mook is a nice piece of research on the interface between plasmonics and magnetism. I will recommend publication provided the authors address the following comments:

We thank the referee for their time and effort and the kind report. Below we address their comments and questions in more detail.

1.) The role of the nonmagnetic ions is not clarified enough. In particular, the authors do not detail how they enter into the calculation, except for the presentation of Eqs. (10) and (11). Please provide the calculation of these quantities accompanied by a drawing to highlight the calculation graphically.

We thank the referee for pointing out that our derivation of Eqs. (10) and (11) was insufficient. We have now updated the manuscript with an additional appendix (App. C), which describes how we obtain these formulas, illustrated with a drawing of the quasi-2D AFM (Fig. 5 in the updated manuscript).

2. Eq. (2) is central to the work. The authors refer to Ref. [26], but there, we only find the same quantity without a derivation. How magnetic moments are coupled to electric fields deserves more than a reference. I urge the authors to provide a derivation of Eq. (2) in the context of the 2D antiferromagnet, as an example. This can be done in an Appendix-

We have taken the comments of the referee to rewrite the text surrounding Eq. (2), offering now more context. Regarding the explicit derivation of Eq. (2), we note that this has been done multiple times in past works (such as PRB **100**, 125161 (2019)), and we do not wish to repeat it here. We have thus chosen to refer to these works in the manuscript.

3. In an Appendix, provide the Feynman rules associated with the diagrams of Fig. 5. Alternatively, provide the corresponding equations of motion.

We thank the referee for this suggestion and now supply the interacting (Matsubara) plasmon Green's function [Eq. (12) in the updated manuscript].

4. The calculations of the quadratic magneto-optic interaction are too sketchy. Provide details in the an Appendix.

We thank the referee for pointing out that our calculations for the quadratic magneto-optic interaction were incomplete. We have addressed their concern by extending the discussion on the magneto-optic interaction, creating Appendix E, which contains a more detailed calculation and additional references.

Given that PRB has no page limitations, there is no reason for the author not to provide a detailed presentation of their work for the benefit of the interested reader.

We hope that updated manuscript is now more complete and offer sufficient details for the interested reader.

Referee B

The manuscript discusses magnon-plasmon coupling and proposes a novel mechanism in which the electric field generated by plasmons is coupled to the electric polarization which appears in an antiferromagnet with the broken inversion symmetry. The mechanism is as far as I am concerned new, and the result deserves publication in some form.

However, before publication can be recommended, I would like to see some changes made which could improve the manuscript.

We thank the referee for their time and effort in reading our manuscript, and for the suggestions they have made. Below we address their concerns in more detail.

My main point is that the explanations are not always clear for the broad audience, and this is why the big picture sometimes gets obscured. Here are the two items which really require putting more accents on.

1. Once the authors write Eqs. (1) and (2), the magnons and plasmons are coupled (assuming the total coupling is not identically zero, which probably can be figured out from the symmetry arguments). Thus, the main point of the manuscript is probably not so much that they are coupled, but that the coupling is strong enough. Indeed, in the introduction the authors dismiss other proposals for magnon-plasmon coupling because they are based on spin-orbit coupling which is weak. In view of these considerations, I would expect to see some discussion elucidating what "weak" and "strong" means, and whether the proposal worked out in the manuscript provides indeed stronger coupling than other proposals do.

We thank the referee for pointing out that we had not defined “weak” and “strong” in our manuscript, and have now done so in the introduction, where we discuss the previous works [third paragraph in Sec. I]. In particular, we no longer use the wording “weak” and “strong”, but instead write “a small coupling rate, such that the splitting is comparable to the damping of the magnon mode”.

We now only use the terms weak/strong coupling in the same context as there are commonly used in the light-matter community:¹

- **Weak** refers to the regime where $g < \{\gamma\mathcal{E}_q, \kappa\}$, where $\gamma\mathcal{E}_q$ and κ are the plasmon and two-magnon mode damping respectively, and g is the effective coupling strength. In this regime, the splitting will not be experimentally observable, because it is overshadowed by the damping in the system. In the results shown in our work, we are never in the regime.
- **Strong** is the regime where $g > \{\gamma\mathcal{E}_q, \kappa\}$, and therefore the splitting is experimentally observable.
- **Ultrastrong** is the regime where $g/\omega_c > 0.1$, with ω_c the energy where the bare bands cross. Here, the splitting is a significant fraction of the bare resonance frequency.
- **Deep Strong** is the regime where $g/\omega_c > 1$, and the coupling exceeds the bare resonance frequency. In this regime, our perturbative approach breaks down.

We have adapted our manuscript accordingly.

2. The authors repeatedly talk about two-magnon continuum, van Hove singularities, and related points, but it is not immediately clear to me how these are related to Figs. 3 and 4. Just to give an example, what does it mean "the plasmon again strongly decays as it enters the Van Hove singularity in the spin-zero two-magnon DOS"? I guess the authors mean that the magnon spectrum is bound from above, and when the plasmon frequency aligns with the upper bound of this spectrum (why is this two-magnon

¹ Kockum et al., “Ultrastrong coupling between light and matter”, Nat. Rev. Phys. 1, 19–40 (2019)

spectrum btw?) the decay strongly increases. However, in the present manuscript this is written almost to the jargonism level and need to be re-written in a more clear language.

We thank the referee for their comments regarding the clarity of the terms used, and we have now rewritten the relevant parts of the manuscript to be clearer.

3. The plasmon, in addition to the electric field (3) also has magnetic field which could couple to the magnetization. This coupling is probably inefficient, but it would be good if the authors could say something about this.

We thank the referee for pointing out this lack of explanation in our manuscript. We had previously discussed the magnetic field of the plasmon and how it would couple to the magnons, realizing a bilinear coupling, in the introduction. We have now also expanded upon this point in Sec. III.A, to explicitly point out that the bilinear coupling can be readily distinguished from the quadratic in magnon operators coupling we consider in this work.

4. I believe the momenta are put wrong in the circle diagram, Fig. 2c.

We thank the referee for pointing this out, we have resolved this issue in the updated manuscript.